1. Python

Introduction to Python

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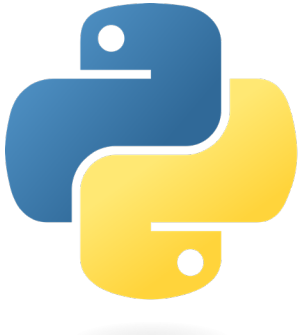
Python

Python is a high-level, interpreted, and object-oriented programming language, known for its simplicity and readability.

For these reasons, Python is an ideal programming language for both beginners and experienced developers.

High-Level Language: High-level languages are English-like and easy to learn and use. The instructions in a high-level programming language are called statements. Here, for example, is a high-level language statement that computes the area of a circle:

$$\mathbf{area = 5 * 5 * 3.1415}$$



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Interpreted Language: An interpreted language uses a program called an *interpreter* that translates a high-level language one statement at a time into machine language and then runs the program. The interpreter will spot several types of errors and terminate the program when one is encountered.

Machine Language: Istructions are in the form of binary code. For example, to add two numbers, you might have to write an instruction in binary code, like this: **1101101010011010**





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Object-Oriented Programming (OPP): It is a programming paradigm that organizes code into **objects**, which are entities that combine **data** (called **attributes**) and **behaviors** (called **methods**), simplifying code organization and making it more modular and reusable.



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The main features of Python include:

- **Simple and intuitive syntax**, which facilitates rapid development of code.
- Support for **imperative, functional, and object-oriented programming paradigms**.
- **Extensive standard libraries**: offering modules and functions for common operations, such as file management, networking, data processing, and much more.

Thanks to its versatility, Python is used in various fields, including web development, data analysis, artificial intelligence, cybersecurity, computational science, and automation.



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2. Programming in Python

Getting started with Python

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Programming in Python

A **code** is a series of instructions that the programmers write. The process of writing a program is also called **coding.**

We are going to create our first code in Phyton:

Create a file named **helloWorld.py**

Use an **editor** to write statements and store them in the helloWord.py file.

```
#My first Python code
print(Hello World!”)

'''This program displays the message:
Hello World!'''
```

Run the code.

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Comments

Comments help programmers to communicate and understand a program.

They are not programming statements and thus are ignored by the interpreter.

In Python, comments are preceded by:

- a pound sign (#) on a line, called a **line comment**,

```
#My first Python code
```

- or enclosed between three consecutive single quotation marks (''') on one or several lines, called a **paragraph comment**.

```
'''This program displays the message:
Hello World!'''
```

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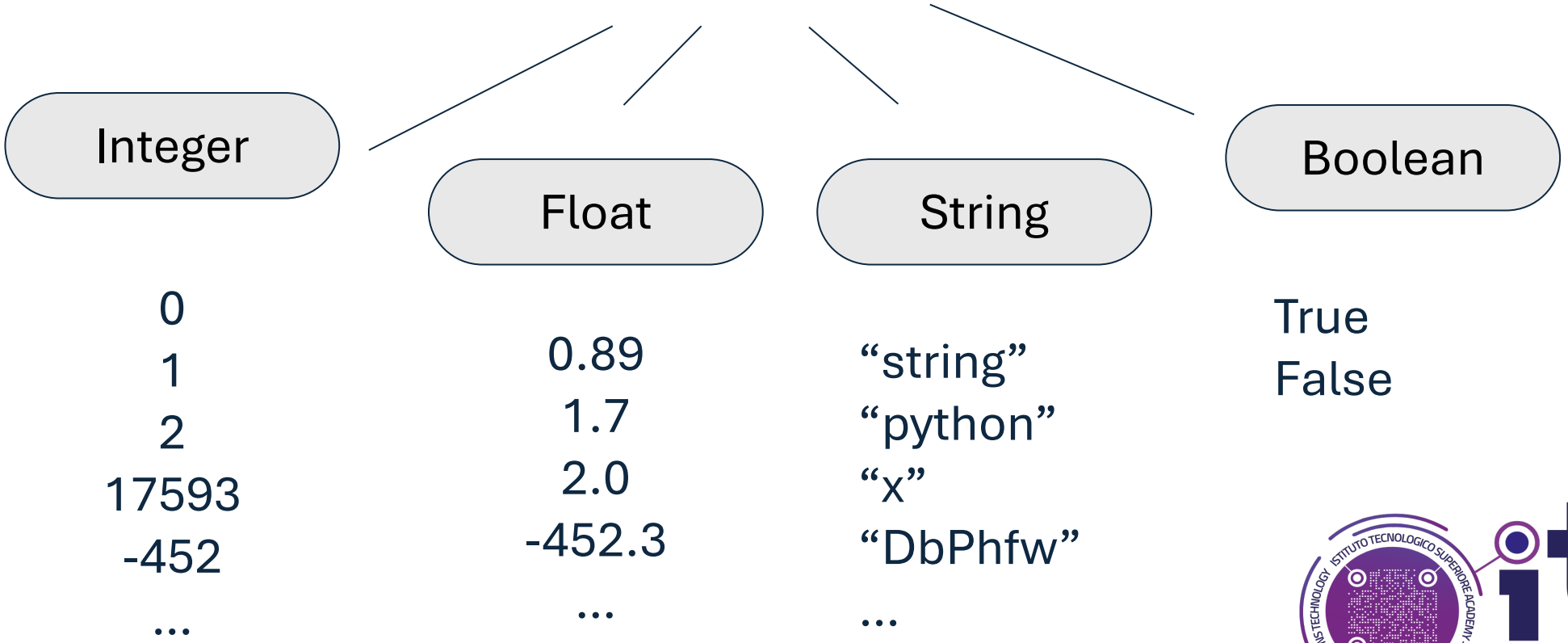
3. Data Types

Primitive Data Types - Storing Information

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What are data types?

A program has to represent different kinds of values to work with.
Let's start with the most basic ones: primitives.



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Data Types

- **Integer (int)**
  Numbers without decimals. Can be positive or negative.

- **Float (float)**
  Numbers with decimals. Can be positive or negative.

- **String (str)**
  Textual data, but really, can be any symbols as long as it's in quotation marks (“ ” or ‘ ’)

- **Boolean (bool)**
  Value that's returned by logical operations. Can only be True or False
  (watch out for case sensitivity!)

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Little Quiz – What's the Data Type?

`123`	Integer
`123.0`	Float
`treehouse`	Error! This is not a String! It's also no other valid data type!
`"89"`	String – Not an Integer
`"False"`	String – Not a Boolean
`true`	Error! Not a Boolean, and not a String! Not a valid data type!

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Variables

Let's first consider the simple problem of computing the area of a circle.
How do we write a program for solving this problem?

Writing a **program** involves designing **algorithms** and then translating them into programming instructions, or **code**.

An algorithm describes how a problem is solved by listing the actions that need to be taken and the order of their execution.

The algorithm for calculating the area of a circle can be described as follows:

1. Get the circle's radius from the user.

2. Compute the area by applying the following formula:

   area = radius * radius * π

3. Display the result.

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Variables

In this problem, the program needs to get the radius value.
The value for the radius can be stored in the computer's memory. In order to access it, the program needs to use a **variable**.

A **variable** is a name that references a value stored in the computer's memory.

Rather than using x and y as variable names, choose **descriptive names**:

In this case, for example, you can use the name **radius** for the variable that references a value for radius of the circle and **area** for the variable that references a value for area of the circle.

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Variables

So we can write our program **computeArea.py**

```
#Assign a value to radius
radius=20

#compute area
area= radius * radius * 3.14

#display result
print(area)
```

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Variables

We want to represent in our program how old we are.

For example:

```
alice = 25

print(alice)
```

Try it out with your own name and age!

Suppose that there's another person in the room named Alice, but her age is different.
We can simply assign a new value

```
alice = 30

print(alice)
```

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Overwriting Variables

We can also modify the data type of our variable. Let's check its type now:

```
type(alice)
```

```
>>> <class 'int'>
```

What if we assign alice a float value?

```
alice = 30.0
```

```
type(alice)
```

```
>>> <class 'float'>
```

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Overwriting: be careful!

Overwriting is easy, but because of this, it's also easy to mess something up.

```
age = 30

name = "Bob"

...

name = 31
```

Later in the code, we wanted to update the age, but accidentally modified the name.

Python won't throw an error here!
Other languages like Java or C++ are typed – for each variable, you predefine its data type. They wouldn't allow this.
Python is more flexible.

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Overwriting Variables

In Python, when a variable is declared, its type is automatically determined based on the value it is assigned.
A variable can have a numeric value:

```
#integerint = -10
print(int)
print(type(int))


#float
flt = 1.234
print(flt)
print(type(flt))
```

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Overwriting Variables

In Python, when a variable is declared, its type is automatically determined based on the value it is assigned.
A variable can have a sequence of characters as its value:

```
#string
string = "This is a string"
print(string)
print(type(string))
```

A variable can have a boolean value:

```
#boolean
bln = True
print(bln)
print(type(bln))
```

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Overwriting Variables

In Python, when a variable is declared, its type is automatically determined based on the value it is assigned.

```
a=2.3
print(a)
print(type(a))

b=10
print(b)
print(type(b))

c='i am a string'
print(c)
print(type(c))

c1="i am a string too"
print(c1)
print(type(c1))
```

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Naming conventions

When naming your variables, there are things you cannot do.

For example: whitespaces

`alices age` is not a valid variable! Instead, e.g.: `alices_age`, `alicesAge`

In general:

- A variable name must start with a letter or the underscore character
- A variable name cannot start with a number
- A variable name can only contain alpha-numeric characters and underscores (A-z, 0-9, and _)
- Variable names are case-sensitive (age, Age and AGE are three different variables)
- A variable name cannot be any of the Python keywords (if, else, import, return, ...)

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Strings

```
#single line string
strA="Hello World"

#multiple line strings
straB="This is too \
    bif to fit \
        on a single line so \
            you multi-lined it"

'''a string is a sequence of characters, so each character of the string can be accessed
using an index
that indicates the position of that character within the given string.'''

name="John Fitz"

print(name[0]) #display character at position 0 of string.

len(name) #return string lenght
print(len(name))

print(name.lower())   # Lowercase
print(name.upper())   # Uppercase
print(name.title())   # Title case
```

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Type Hinting in Python

Integer

```
num:int = 3
```

Float

```
numf:float = 4.5
```

String

```
message:str = "Hello World"
```

Boolean

```
flag:bool = True
```

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